Translator Reader User Guide

Read the original, translate in place, and keep your notes.

Abstract: This English guide is also the sample document. Import it to practice translation, selection, image OCR and export. Before configuring a translator, read docs/USER_GUIDE.zh-CN.md.

Keywords: PDF reading, translation memory, captions, annotation, local storage

1. Quick Start

1. Install Python 3.11+ and Node.js 22.12+. Run python scripts/setup.py from the repository root.

2. Run python scripts/start.py. Open http://127.0.0.1:8000 and use the User Guide button or import a PDF.

3. Wait for parsing. The default mock provider demonstrates the workflow; it does not produce a real translation.

4. Choose your provider in Settings, then translate the document. Failed segments can be retried.

2. Configure Translation

For a local Ollama service, choose the OpenAI-compatible provider and Base URL http://127.0.0.1:11434/v1. Use the exact installed model name.

Choose simple single-segment mode for translation-specialized models. Use numbered batches for instruction models that preserve the output markers.

For a cloud service, set its Base URL and model in Settings. Put the key in .env under the configured API key environment variable. Restart after editing .env; never commit it.

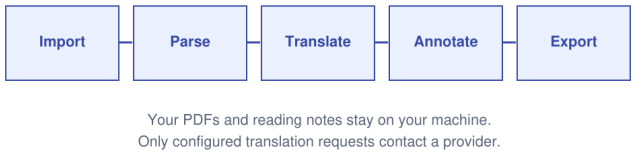


Fig. 1: The reading workflow. Try translating the labels and this caption.

A safe first exercise

Import this guide, translate it, select one complete sentence, add a note, and export the annotated PDF. Check both the marker on the original page and the notes appendix.

Automatic translation is optional. Parsing and local OCR do not require an API key. Real translation uses your configured service and may incur provider charges.

Installation downloads dependencies. Once installed, parsing, OCR, annotation and export run locally. A remote translation or vision service receives the text or image you ask it to process.

Read precisely and keep useful notes

3. Reading and Selection

The left pane is the source PDF. The right pane uses the same pages with translated text at the original positions. Titles keep source size and weight where space permits; figure graphics, table cells and formulas are preserved.

Use dual-pane, source-only or translation-only view. Scroll positions stay synchronized. Zoom reuses page content to reduce work on long papers.

Drag from the first character to the last character of a sentence. Selection stays in its starting column; drag near the viewport edge to scroll. Double-click selects a word; triple-click selects a paragraph.

Click a paragraph for copy or retranslation actions. Ctrl+F searches the current paper; library search scans all papers. The outline lists detected headings.

Keyboard shortcuts

J / K or PageDown / PageUp: next / previous page. Ctrl+Home / End: first / last page. + / - / 0: zoom / fit width. 1 / 2 / 3: dual / source / translation view. ?: help.

Ctrl+wheel zooms at the pointer. H toggles the hand tool; Space+left drag or middle drag pans the page. Ctrl+Z undoes saved note/highlight edits; Ctrl+Shift+Z or Ctrl+Y redoes. Text fields keep native undo. History ends when you leave the paper.

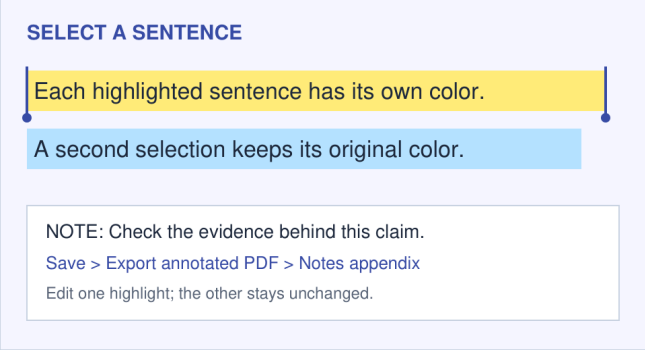


Fig. 2: Precise selection, independent colors and a saved note. The image shows an example, not interactive controls.

4. Highlights, Notes and Export

After selecting text, choose Highlight or Highlight and Note. Each selection is independent: editing one color does not recolor other selections.

Save with Done or Ctrl+Enter. Esc also saves. If saving fails, the draft remains. Use the annotation list to edit, delete or locate selections.

Reading notes can quote a paragraph or refer to a page or the whole document. Page markers open their notes. Wait for saves to finish before exporting.

Table 1: What each export contains.

Format	Contents
Markdown	Source, translation or bilingual text, plus reading notes.
Annotated PDF	Original PDF, highlights, markers and a visible notes appendix.
JSON	Blocks, translations, annotations and notes; no one-click restore UI.

PDF export contains original pages with notes, not the translated browser overlay. Use Markdown for translated text. Some viewers hide popup comments; the visible notes appendix contains their text.

Translate figures and reuse your work

5. Images and Captions

Translate Image and Caption reads PDF text first, then uses local RapidOCR / PP-OCRv4 for raster text. Recognized labels go to your text translation provider.

The same action recovers missing captions on older documents and translates the nearest figure caption, even outside the image crop. Full-document translation also recovers native captions without re-parsing.

Detected raster-only captions are OCR candidates. Blurred text and low-confidence labels may remain untranslated for review. Table captions are translated separately from table cells.

Use the correction dialog to inspect recognized text. Draw a region for OCR or choose Vision Recognition when needed. Its separately configured service must accept images. A default model name does not install or start that model.

Layout detection

DocLayout-YOLO is optional. Install its package and DocStructBench weights using docs/INSTALL.md for figure, table and caption localization. Without them the reader uses heuristics. Whole-page scanned-paper OCR is not yet a complete import pipeline.

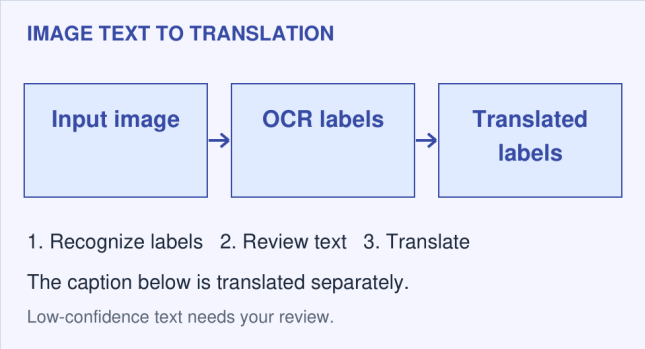


Fig. 3: Raster labels for OCR practice. Translate this image and its caption, then review the recognized words.

6. Translation Memory and Data

Successful real translations are reused for matching source text, target language and forced glossary. Mock output is excluded. Changed text or glossary can require new requests; re-parsing is not always free.

Continue Translation fills missing work. Force Retranslate bypasses reusable results and requests the current provider. Repeated image translation reuses successful matching captions.

Re-parsing clears current paragraph translations. Notes remain, with quote-based reattachment where possible. Prefer automatic caption recovery when only captions are missing.

PDFs, SQLite records and settings live in data/, or APP_DATA_DIR. JSON export archives a paper but is not a complete database backup. Stop the app before copying the entire data directory; protect .env separately.

Privacy and access

Do not publish private papers, app.db, API keys or pairing tokens. LAN mode is explicit: python scripts/start.py --lan. Use the pairing link on trusted devices. Keep one server worker; this is a personal reader.

Practice formula: H = N_hit / N_total × 100%. This is a symbolic example, not a benchmark. Verify important equations and numbers against the source.

Troubleshoot, contribute and practice

7. Troubleshooting

Blank interface: complete setup and rebuild the frontend. /api/health checks the backend only. A missing frontend build reports how to generate it.

Translation fails: check model, Base URL, protocol and API key variable. Review the error and retry. A local model must already be running.

Caption missing: try Translate Image and Caption. If a raster caption was not detected, crop it for OCR or vision and review the result.

Note missing in a PDF viewer: open the visible notes appendix at the end. Export again after saving your latest edits.

8. For Developers and Agents

Read README.md, AGENTS.md and docs/INSTALL.md. Tests use temporary data and mock services. Run python scripts/run_tests.py, build the frontend, then run npm --prefix frontend run test:browser. Do not modify an existing library or use paid translations as a test.

References

[1] README.md and docs/USER_GUIDE.zh-CN.md: setup and current operation instructions.

[2] PyMuPDF: pymupdf.readthedocs.io

[3] DocLayout-YOLO: github.com/opendatalab/DocLayout-YOLO

[4] RapidOCR: github.com/RapidAI/RapidOCR

[5] Ollama: docs.ollama.com/api/openai-compatibility

Appendix A: Practice Checklist

Select this sentence and add a note about your reading goal. Highlight another sentence in a different color. Edit the first color and confirm the second stays unchanged.

Create a page note. Export Markdown and the annotated PDF, then inspect their note sections. Check the figure and table captions on earlier pages after translation.

Switch views and zoom several times. Reopen the paper to check its remembered page. References above remain untranslated; this appendix should still be translated.